PAPER_1272_VACUUM_STABILITY

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PAPER_1272 — Higgs Vacuum Stability via Static Ledger Closure

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: B — Physics Foundations (CLOSED)
Date: June 11, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — UQFF integer-primitive derivation
Calculator surface: `calculate_paradox({"paradox": "vacuum_stability"})`
Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1272})`

Master Expression

```
w = -1 exact + F_U = 1 → vacuum stable by construction (no false vacuum decay)
```

Match: STRUCTURAL EXACT

Physical Interpretation

The closed static vacuum ledger guarantees vacuum stability — there is no lower-energy state to tunnel to. False-vacuum below Planck excluded.

UQFF Locked Primitives

- ρ_SCm = 7.09 × 10■^{3}■ J/m^3 (vacuum density)
- S_26 = 1.453162, S_26_DPM = 1.4531 × 10^2■
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84
- F_TRZ = 0.1, Λ = 0.00729735 (= α fine-structure)
- ω_SCm = 1.25 THz, λ_i = 1.0
- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "vacuum_stability"})["value"]
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_vacuum_stability_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["vacuum_stability"]`
